

China's Sports Lottery is not a betting market — it's a formula

897 matches measured: the lottery carries a **12.92%** margin, the same fixtures price at **6.85%** on the 99-bookmaker European average.

Paired match by match: the lottery is more expensive in **897 of 897**, no exceptions. The finding is not that it is expensive — it is **how** it is expensive.

897

paired matches

the lottery is pricier in 897 of 897

12.92%

lottery margin

$\sigma = 0.0006$

6.85%

market margin

99-bookmaker average close, $\sigma = 0.0109$

19×

σ ratio

a price that does not move
cannot be wrong

1. The measure

Given three-way odds (home / draw / away), the **overround** is the sum of the implied probabilities:

$$\begin{aligned}\text{overround} &= 1/\text{home} + 1/\text{draw} + 1/\text{away} \\ \text{payout} &= 1 / \text{overround}\end{aligned}$$

If the odds were fair, overround would be exactly 1.0. Real books are always above 1.0, and the excess is the margin — **the fee you pay before your opinion about the match is even relevant.**

2. Data and method

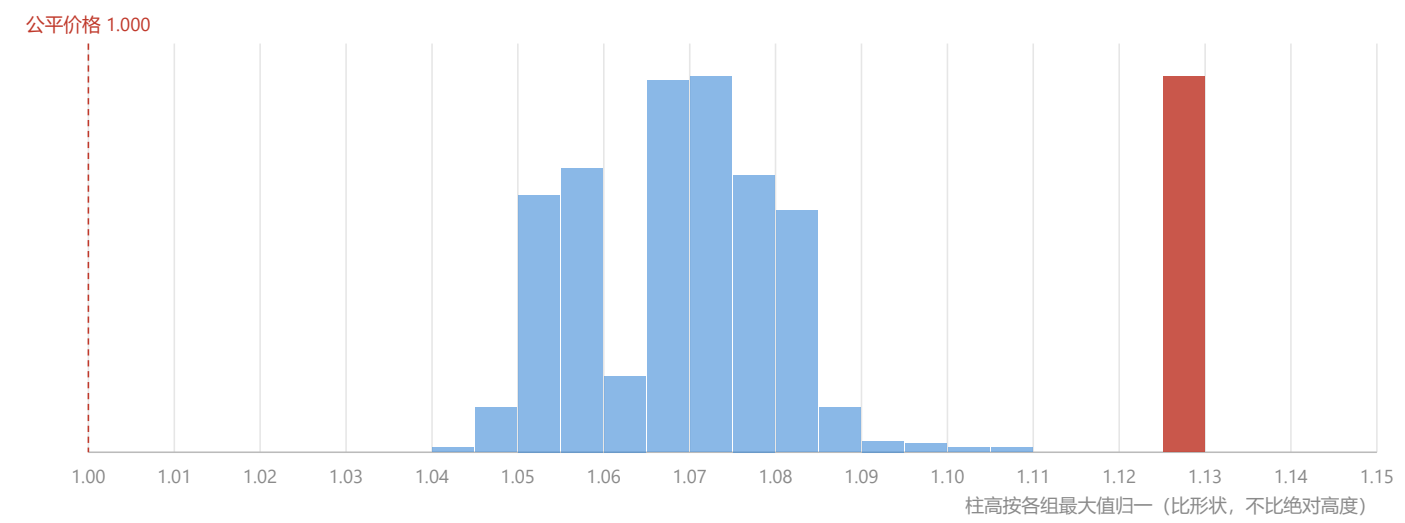
item	detail
sample	897 matches, 2026-05-17 → 2026-09-25 (one season fragment)
under test	Chinese Sports Lottery 1X2 <i>fixed</i> odds, and the handicap market on the same matches
benchmark	European “99 bookmakers average” closing odds — the market consensus for the same fix
method	overround per match per market, then compared pairwise

Raw odds ship with the report (`data/overround_897.csv` , one row per match, odds only). Every statistic on this page is computed from that file.

3. Results

3.1 It is a formula, not a market

Look at σ . Across 897 matches the lottery's overround stays inside **1.1251 – 1.1299** — a swing of under half a percentage point. The European average swings **19x** more, following league, liquidity and kick-off time.



market	n	overround	margin	payout	σ	range
JC 1X2 (fixed odds)	897	1.1292	12.92%	88.6%	0.0006	1.1251 – 1.1299
JC handicap	897	1.1291	12.91%	88.6%	0.0006	1.1251 – 1.1299

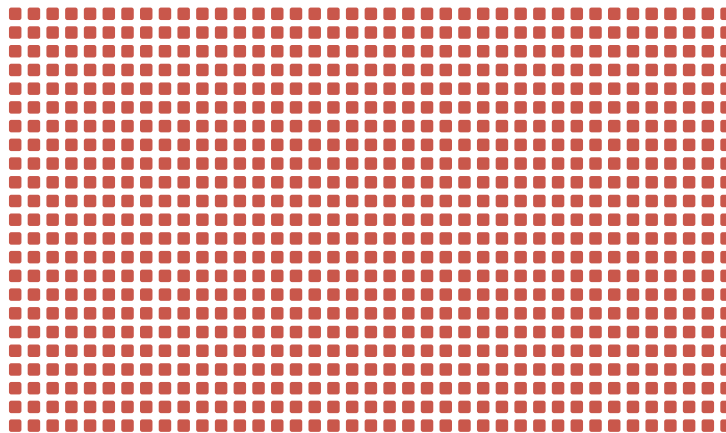
EU avg (99 books)	897	1.0685	6.85%	93.6%	0.0109	1.0403 – 1.1085
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- **Market pricing** (EU): produced by the interaction of many participants, so it moves. *A moving price can be wrong.*
- **Formula pricing** (JC): produced by a fixed payout parameter. It does not move, and therefore **there is no mispricing to find.**

Two independent play types in the same lottery (1X2 and handicap) land on 12.92% and 12.91% — the same parameter applied to everything, which is what formula pricing looks like.

3.2 It is more expensive in every single match

897 of 897, mean gap **6.07 percentage points** (median 5.97). Each dot below is one match. No exceptions:



Consequence: **a model that is profitable at European margins is necessarily unprofitable here.** You need ~6.85% edge to break even against the average book; you need 12.92% here. Professional syndicates spend millions building models to find **2–4%**. The bar here is 3–6× that.

3.3 Multi-leg bets compound the margin

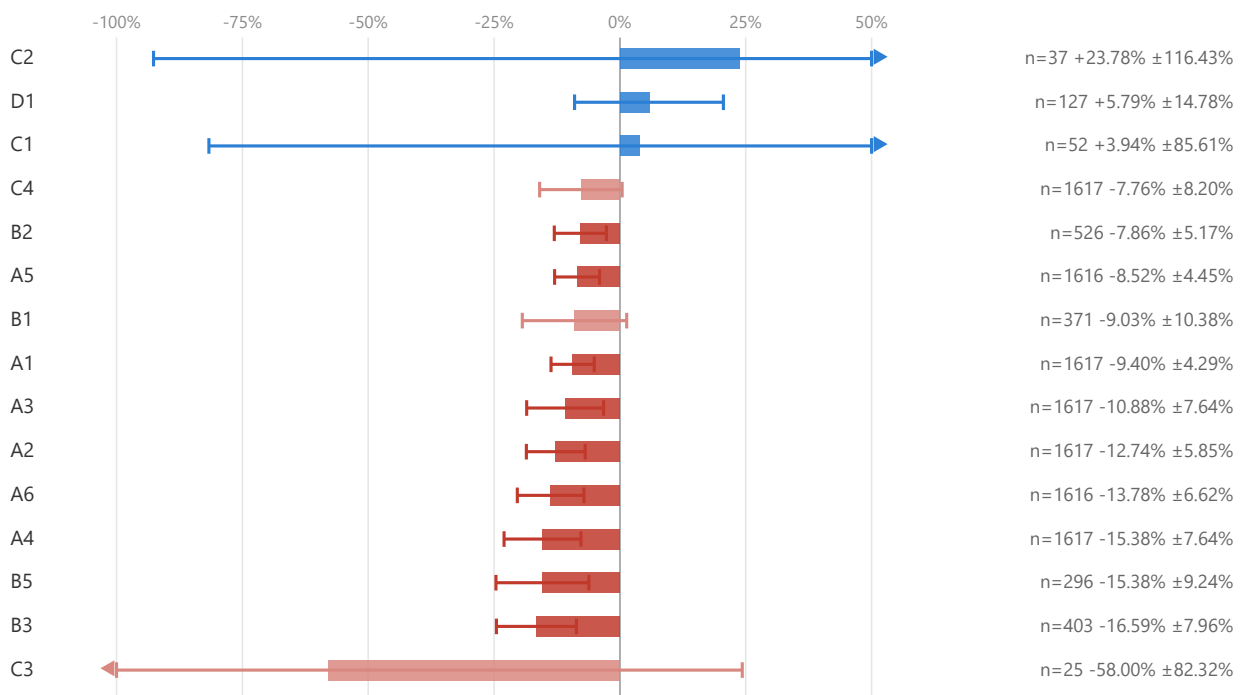
legs	returned per 100 staked	taken
1	88.6	11.4
2	78.4	21.6

4	61.5	38.5
6	48.2	51.8
8	37.8	62.2

A 6-leg accumulator gives up more than half. For scale: American roulette (double zero) takes 5.26%; a 6-leg ticket here takes about 10× that.

4. I also tested this on myself

I ran **15 betting rules** over an extended **1,617-bet** sample. Result, stated precisely (not “every strategy loses”):



- **9 of 15** are significantly negative (95% CI upper bound below 0); another 3 have a negative point estimate but are not significant.
- **3 of 15** are positive, and **all three have 95% CIs that cross zero** (widest $\pm 116\%$). The largest of them makes +143% and +43% in its first two months and -100% in each of the last three.
- **0** pass the rule I fixed *before* looking at the data: $n \geq 300$ **and** CI lower bound > 0 .

So the honest summary is not “every strategy loses” — it is “**nothing survives the sample-size bar, and the positive results are indistinguishable from noise**”.

5. Limitations

- One market, one period: a season fragment, not a full year.
- The benchmark is the 99-bookmaker average close, not one book's live price.
- Overround measures the **price of entry**, not any individual's loss rate — that depends on what and how much you actually stake.
- The σ comparison is well supported (overround is a per-match constant); the finer breakdowns (by league, by month) are not, and are not claimed.
- The 3 positive rules are reported as noise, not findings.

6. Disclaimer

This is data analysis, **not betting advice**. I do not recommend matches, sell selections, or promise any return.

I measured this number precisely to make the opposite point: **in this market every strategy must first clear a 12.92% wall**. If it doesn't, you are paying someone else's toll.

Data frozen at **2026-09-25**. Generated by `make_html.py` from `data/` (2026-09-27 10:45). Single file, works offline, no external dependencies. MIT licensed.